

Group work set 1: Basics

Q1. Imagine a town in the mountains, where the local economy thrives on the production and sale of “widgets”. The townsfolk rely on these widgets for various tasks, from household chores to powering machinery in the local factories. The townsfolk are reluctant to use banking services, so they hide cash inside their houses. One day, a large fire hits the town, burning 50% of the houses in town. In response, the government asks residents to donate money for repairing the harm caused to the city hall, which also burnt down. Alas, very few people donate. Hence, the government decides instead to impose a per-unit tax on the production of widgets to fund the repairs. Meanwhile, a group of young engineers unveils a groundbreaking invention—a new technology that revolutionizes the way widgets are produced. This innovation slashes production costs dramatically, making widget manufacturing more efficient and affordable than ever before.

Explain what is the anticipated effect of these events on the equilibrium price and quantity of widgets in the market. Provide a brief rationale for your answer.

Q2. "GreenTech" is a firm that produces solar panels. GreenTech has been enjoying high profits for the past year and therefore decided to donate money to the election campaign of the minister of finance. However, two changes have occurred in the market: First, a breakthrough in the production of silicon, a key material used in the production of solar panels, has made it cheaper and more readily available. Second, a new government policy has been implemented, which required the installation of solar panels in every governmental office.

How should these events affect the equilibrium price and equilibrium quantity?

Q3. Consider a market for coffee with regular supply and demand curves. The equilibrium price is \$3 and the equilibrium quantity is 1,000 units. Suppose that two events occur simultaneously: First, the price of tea increases. Second, a new roasting technology is developed that lowers the cost of producing coffee. According the model covered in the course, does the equilibrium price increase, decrease, or stay the same? Does the equilibrium quantity increase, decrease, or stay the same? Explain why.

Q4. Maskovid is a firm that produces facial mask during the Covid-19 pandemic. The market is competitive, i.e. the firm is not a monopoly and there are regular supply and demand curves. Assume that the law says that facial masks must initially be worn indoors by every person. Then, one day, the government announces the following change in policy: From now on, wearing masks indoors is no longer mandatory. Furthermore, the government will subsidize the manufacturing of masks, by giving producers 1\$ per mask produced. Discuss what would be the possible effects of this chain of events on (1) the equilibrium price and quantity of facial masks, (2) the consumer surplus and the supplier surplus, and (3) efficiency?